

ELCT918 – Lab 3 Report

Real-Time Handwritten Digit Recognition: From a Trained CNN to a Webcam OCR Pipeline

Team members

Name	ID
Nada Yehia Elsayed	S2701048
Sama Samy Mohamed	S2701051

Instructors:

Dr.Eman Azab

Eng. Nour Elshahawy

1. Task 1

Test accuracy	98.52%
Number of parameters	61,750
Training time per epoch	Epoch 1: 10.74 seconds Epoch 2: 2.68 seconds Epoch 3: 1.70 seconds Epoch 4: 2.15 seconds Epoch 5: 2.16 seconds Epoch 6: 1.73 seconds Epoch 7: 1.73 seconds Epoch 8: 1.77 seconds Epoch 9: 1.76 seconds Epoch 10: 1.93 seconds

2. Task 2

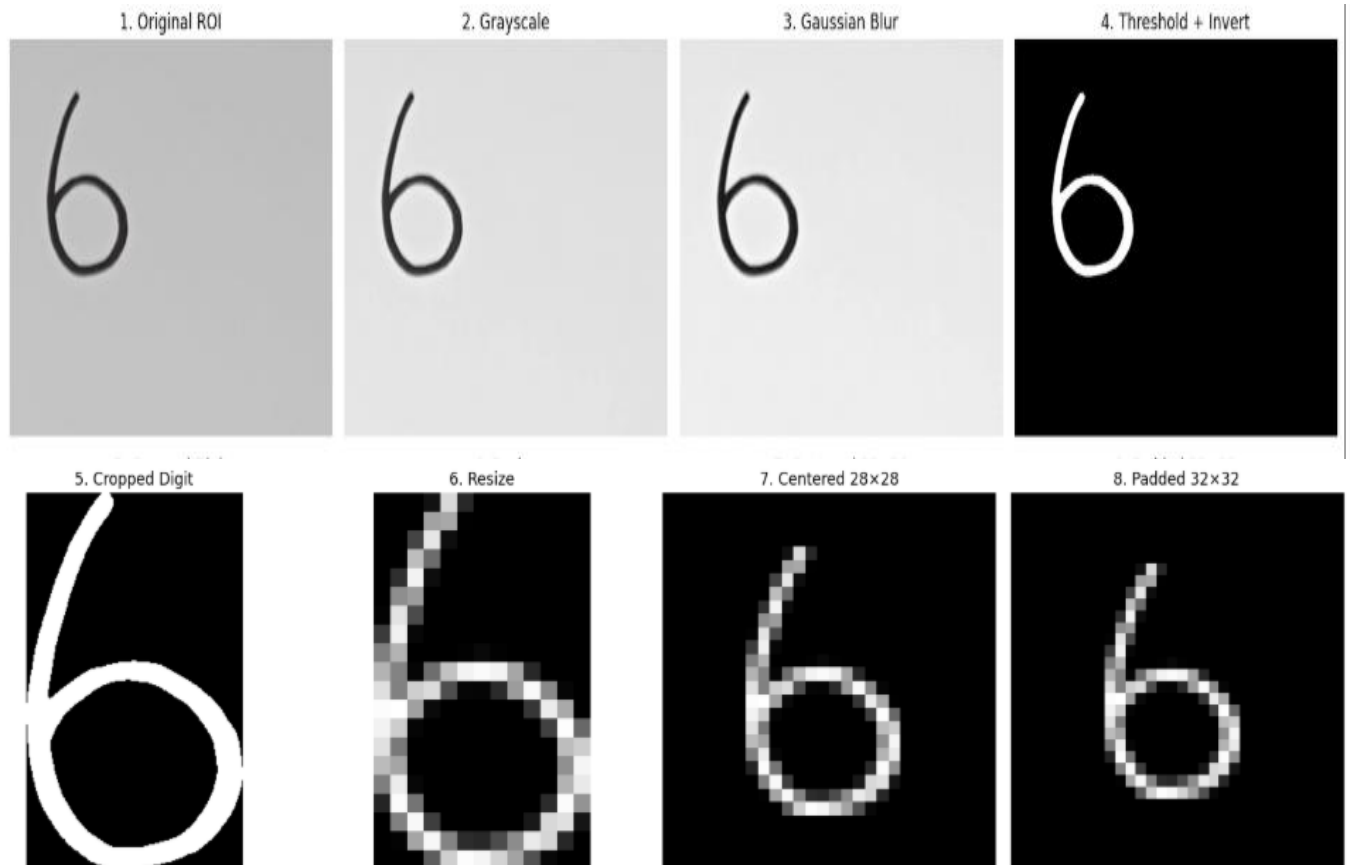
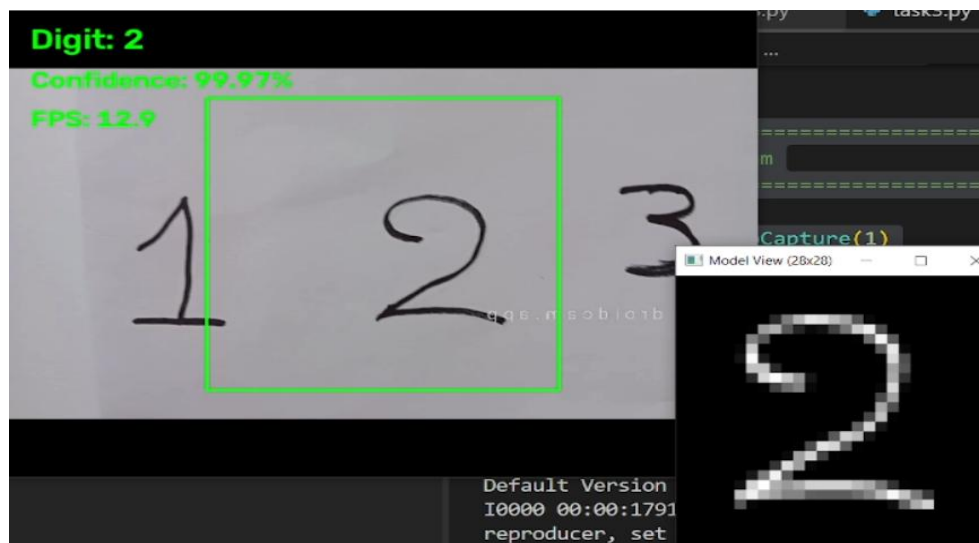
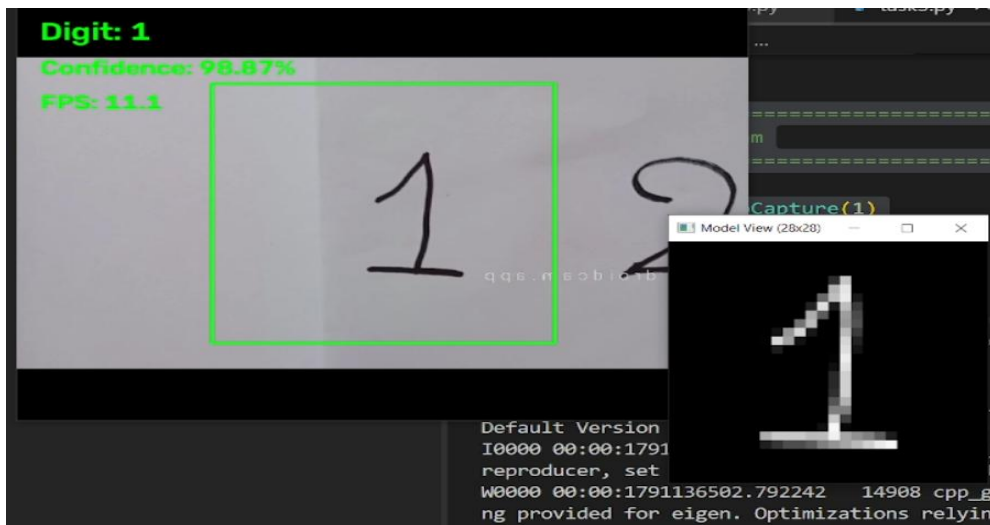
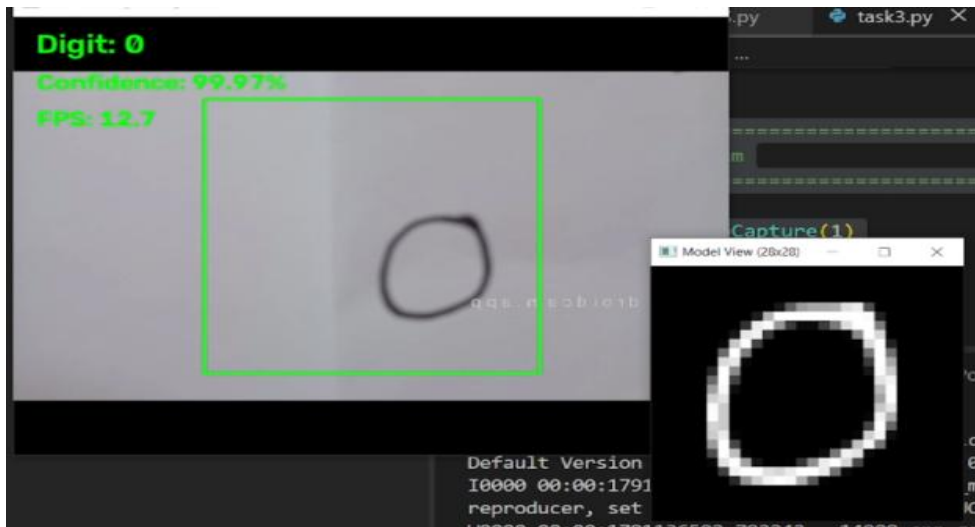
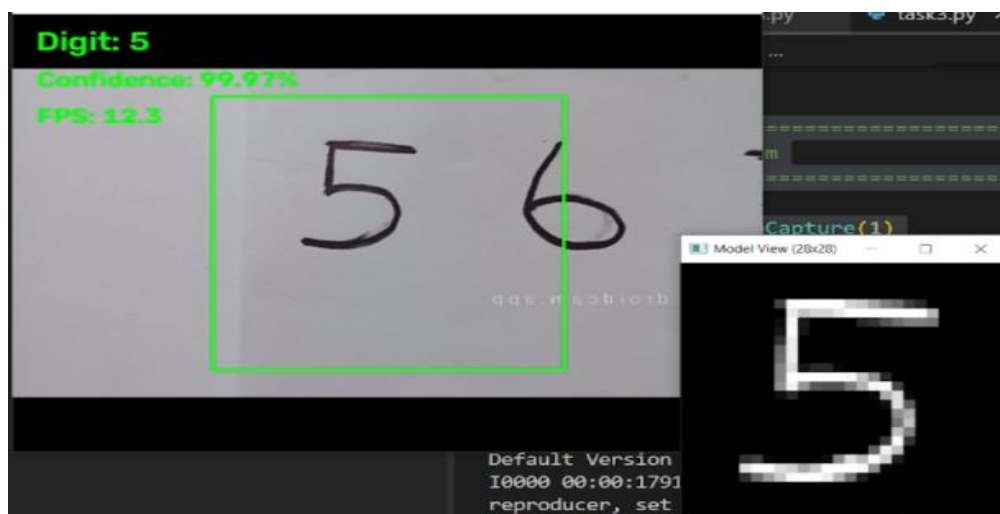
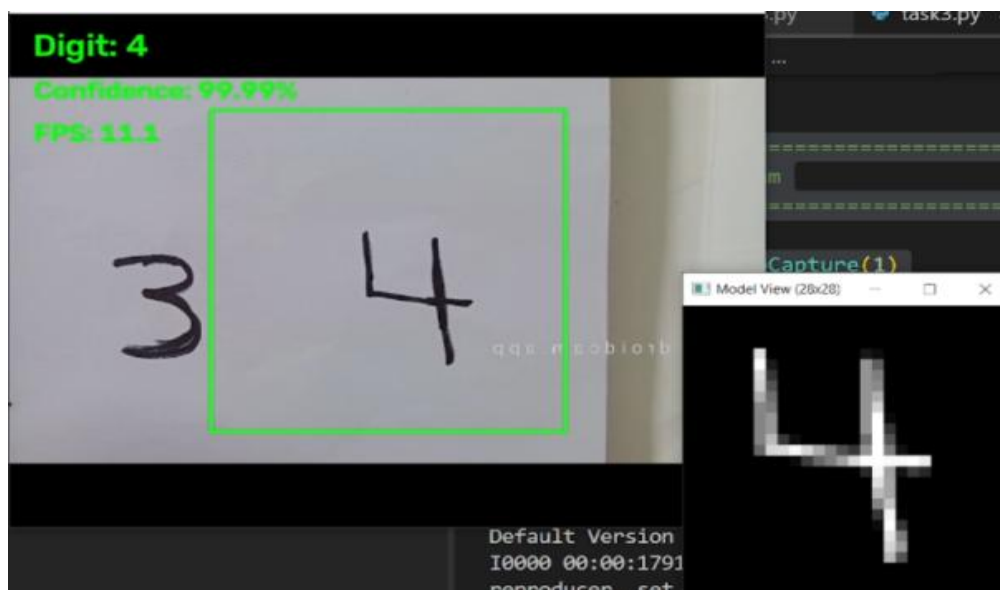
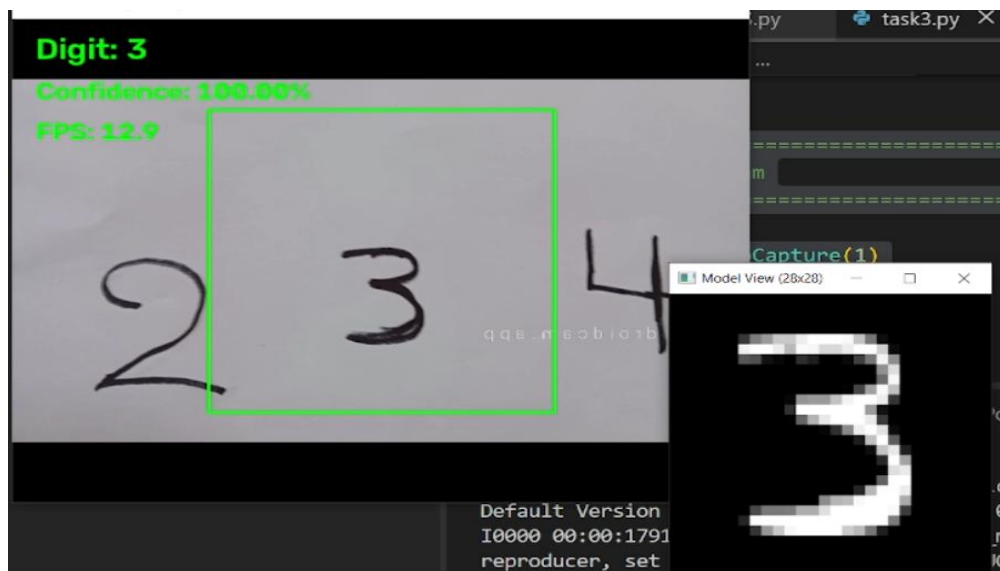
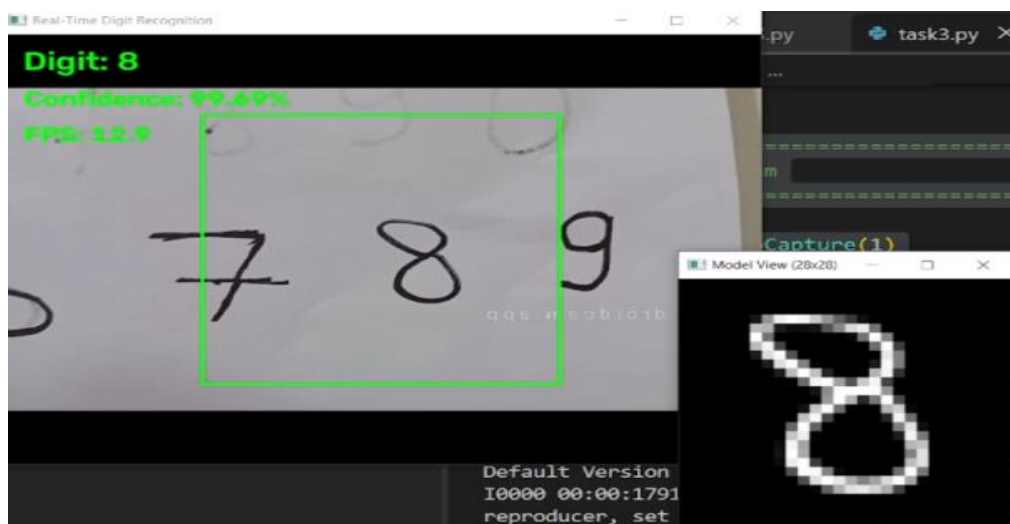
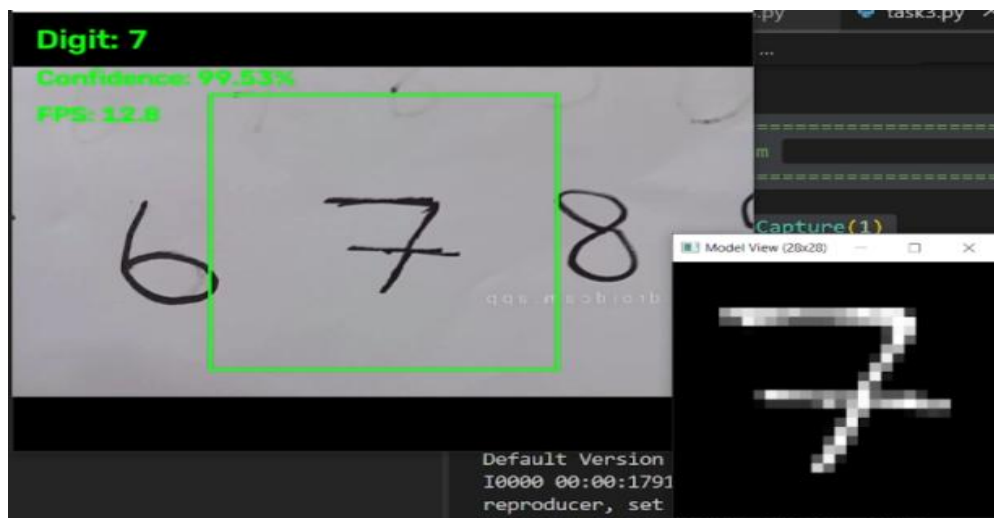
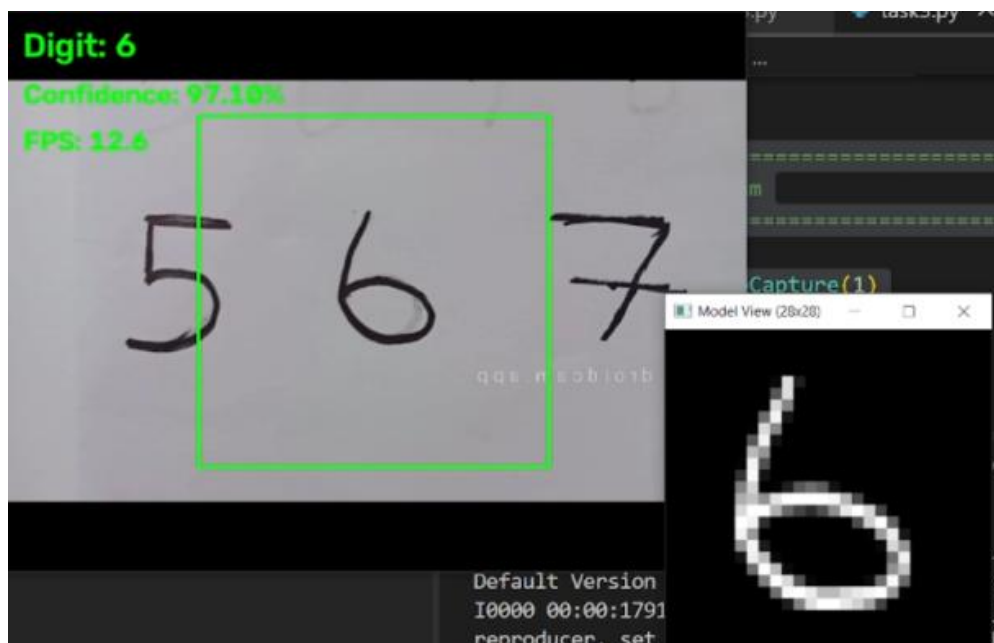


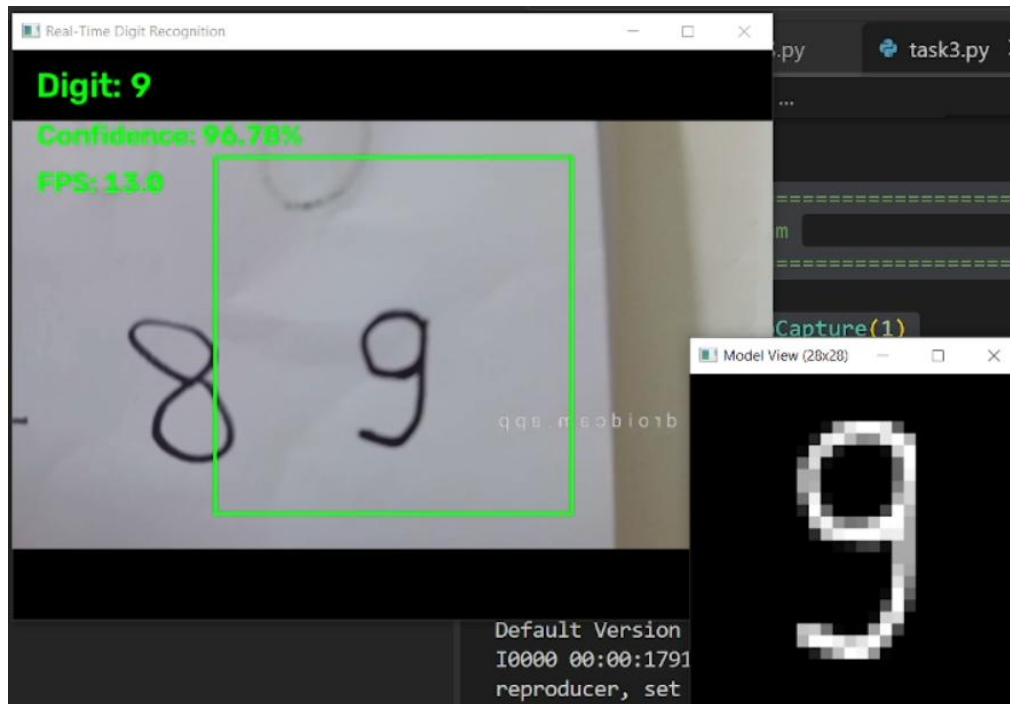
Figure 1: preprocessing steps

3. Task 3



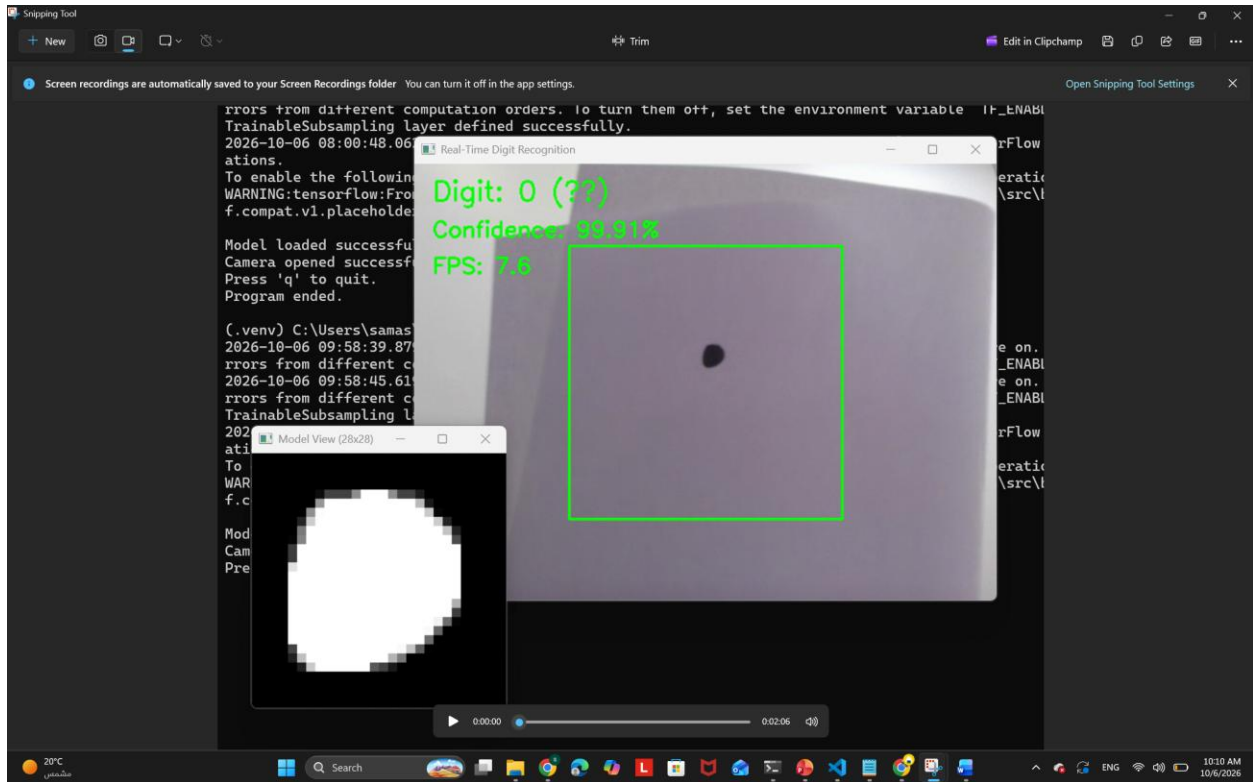


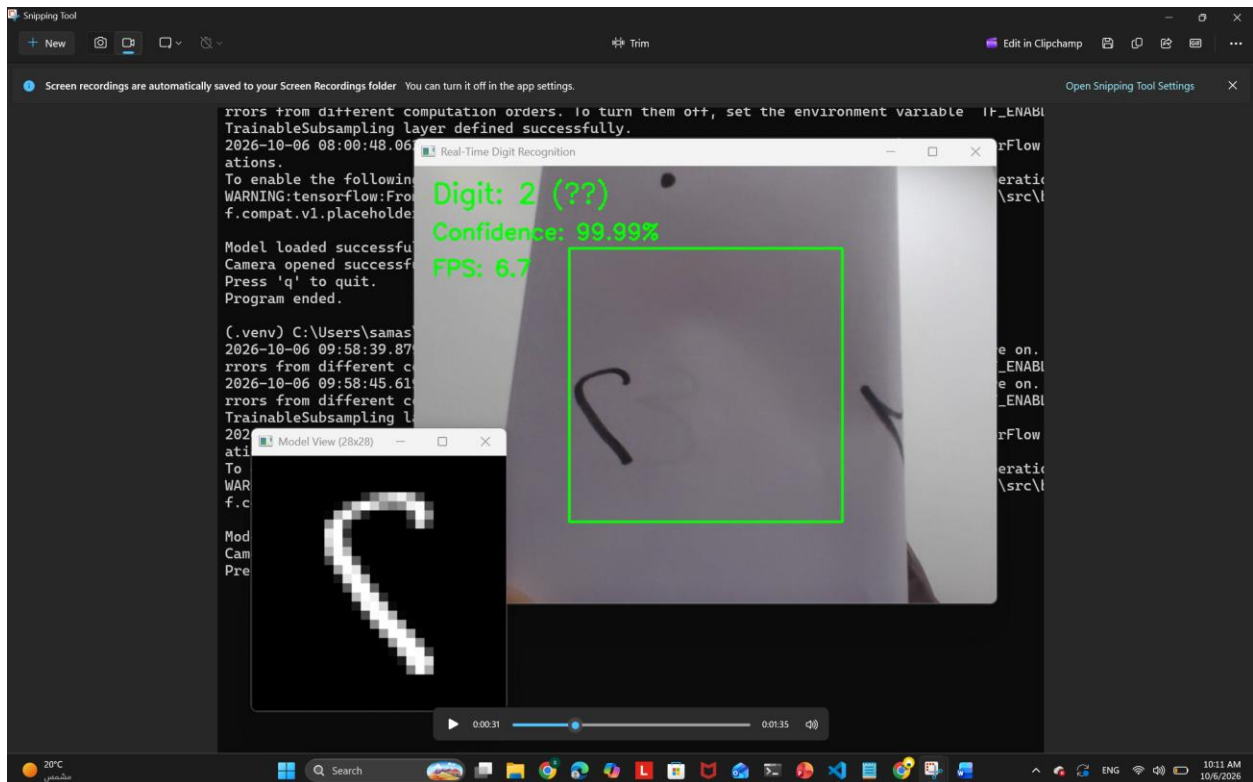
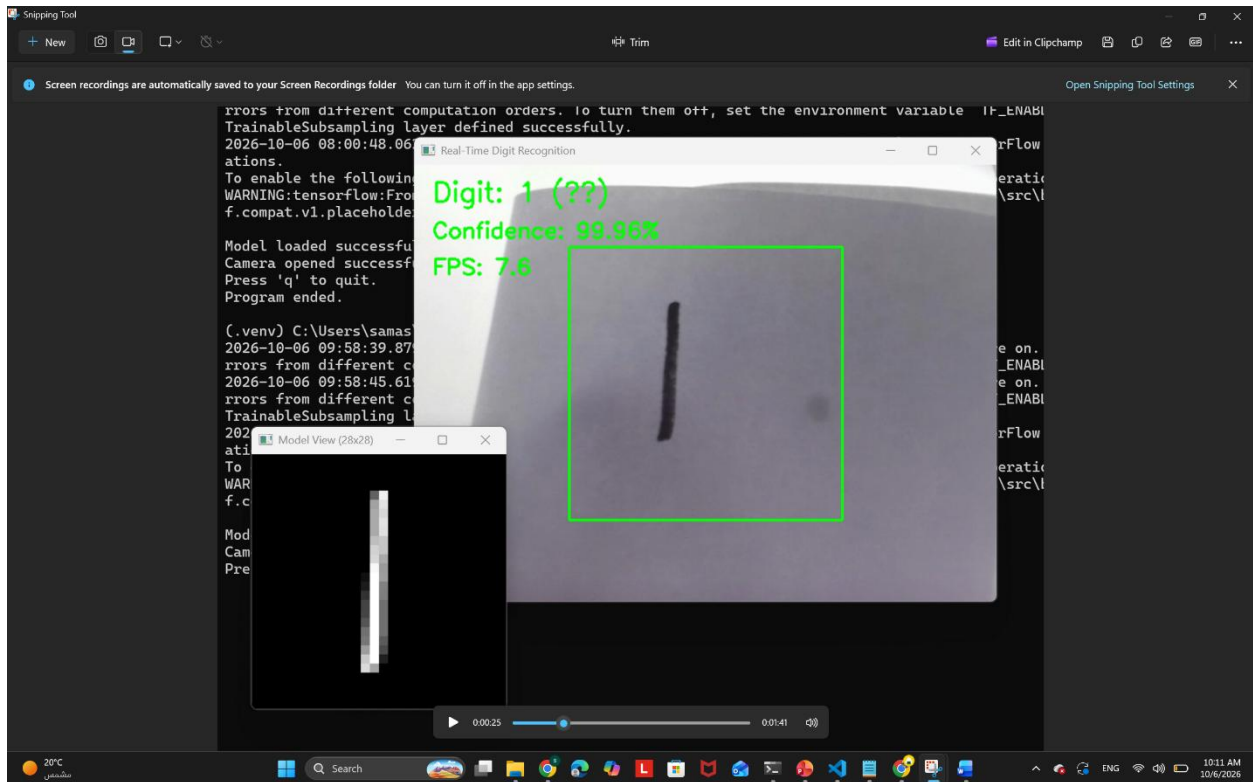


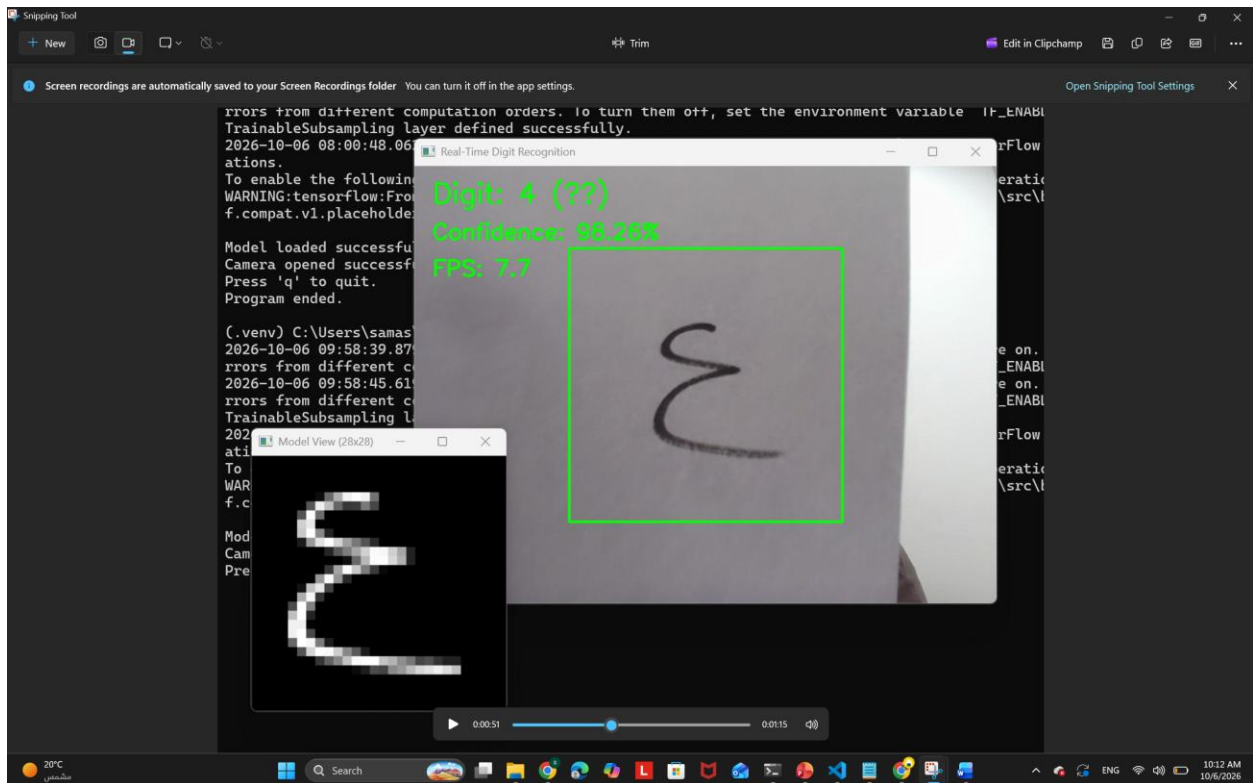
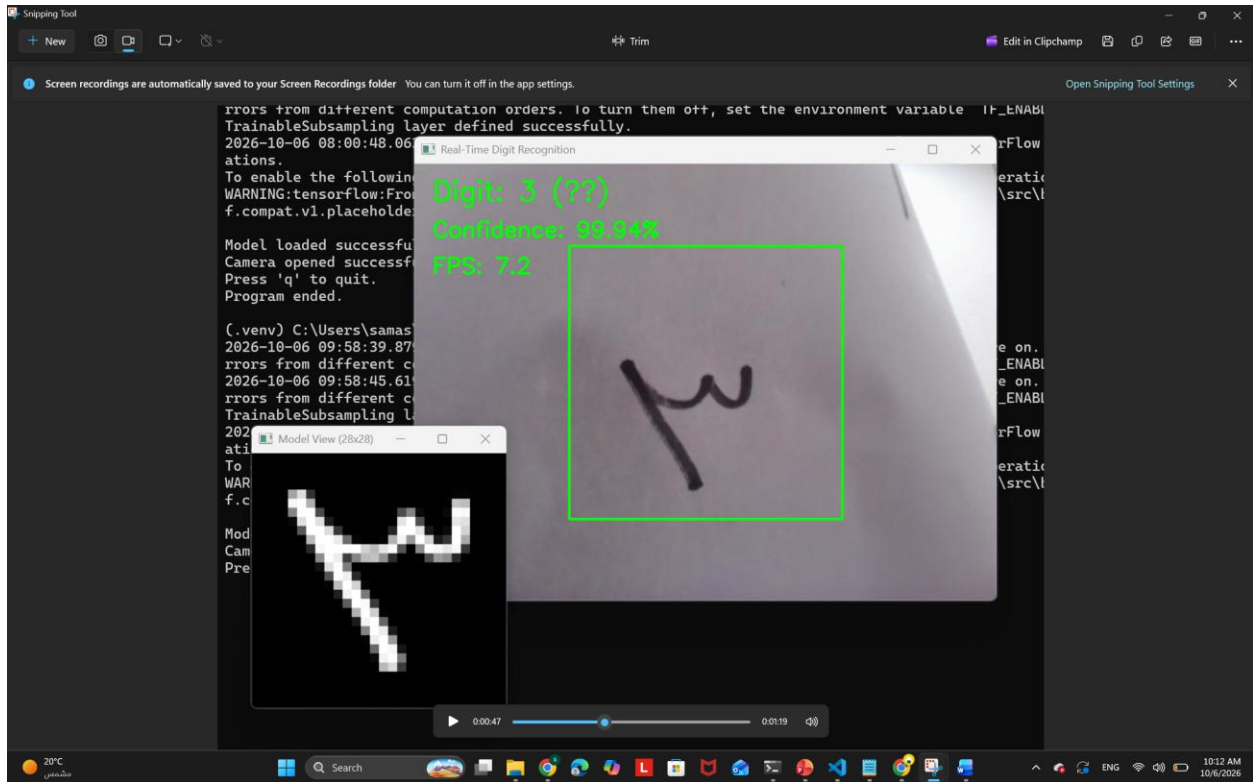


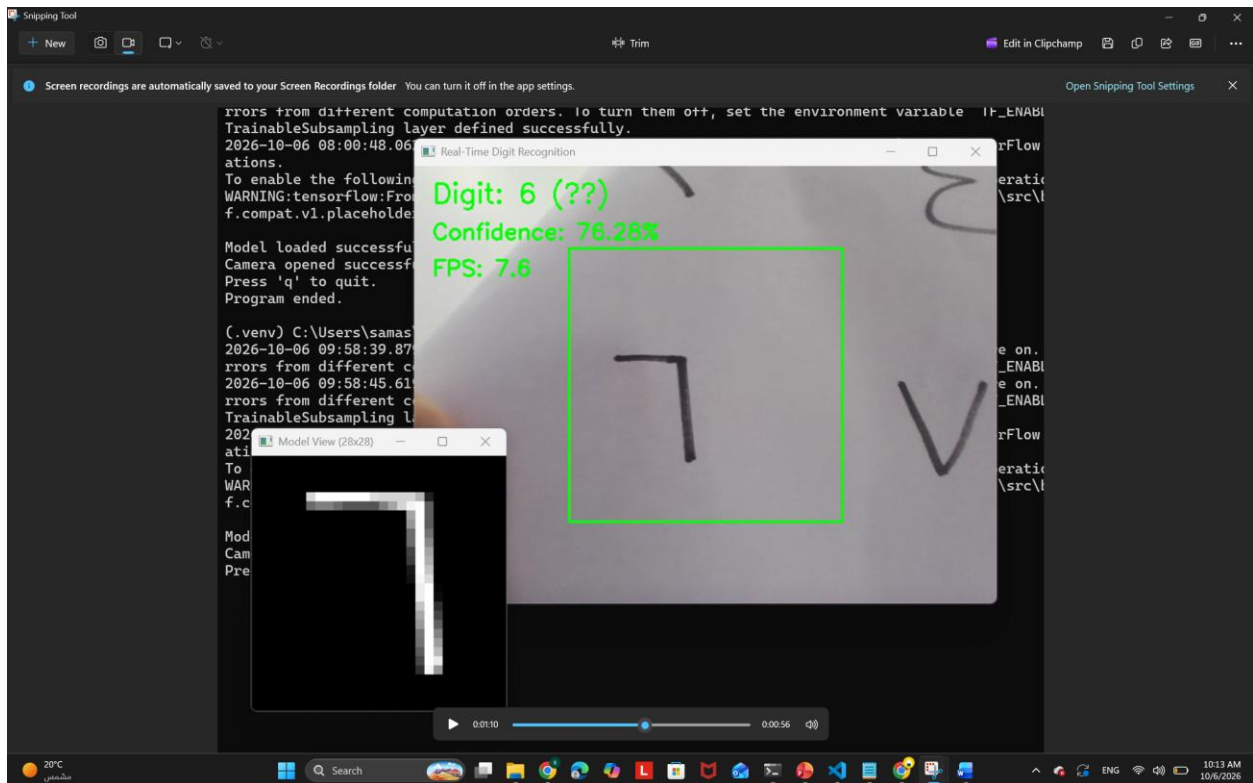
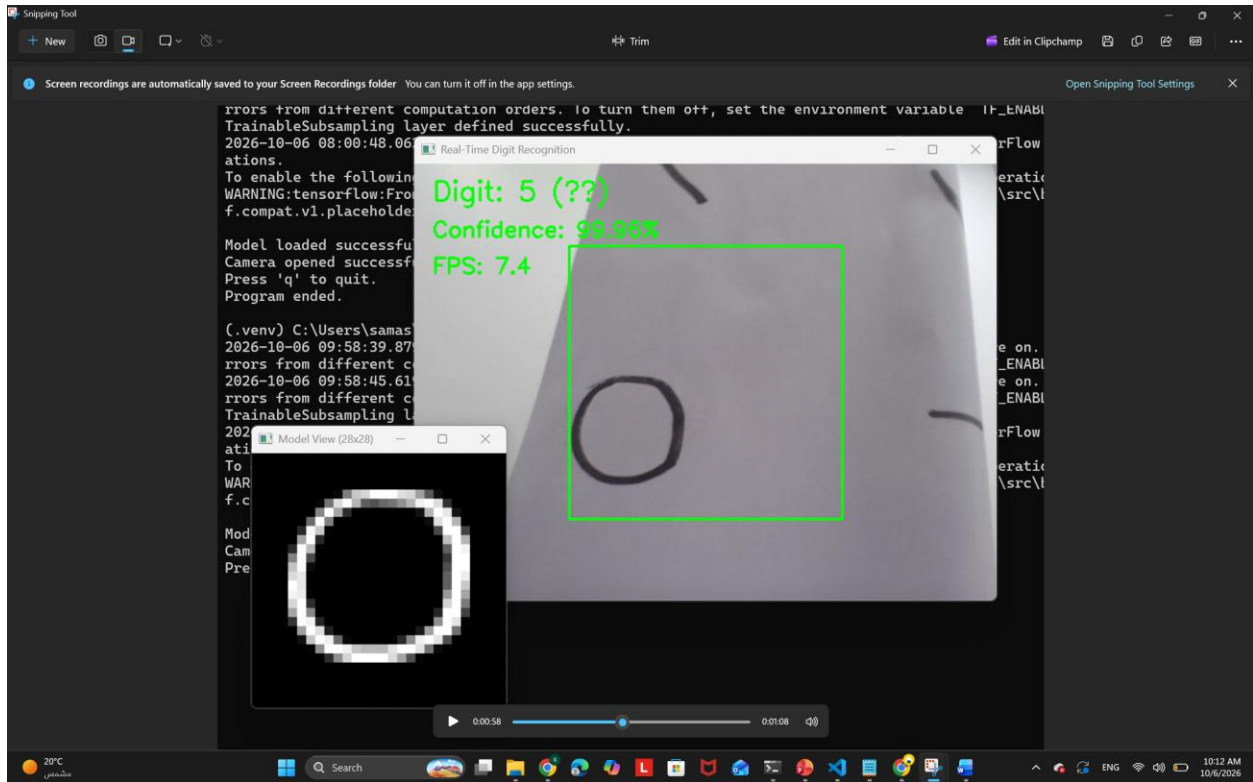
The model recognized all the tested handwritten digits correctly without any noticeable confusion between digits.

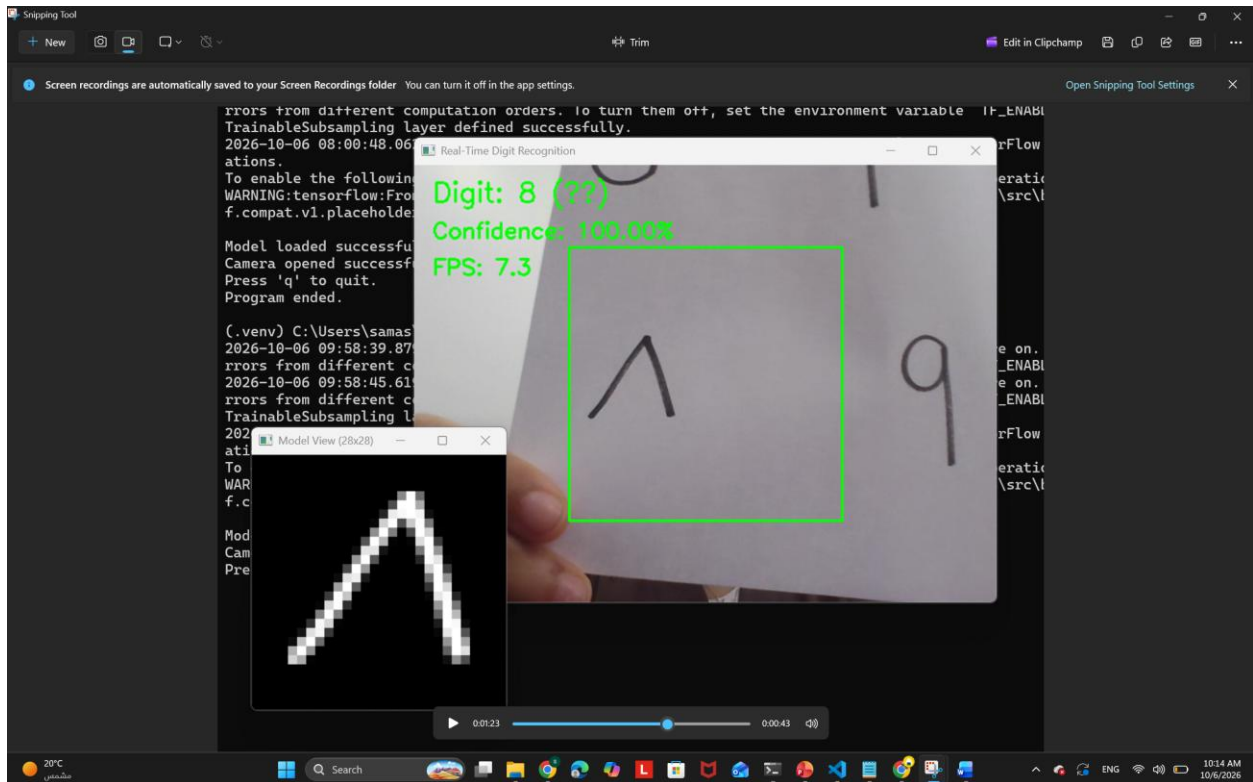
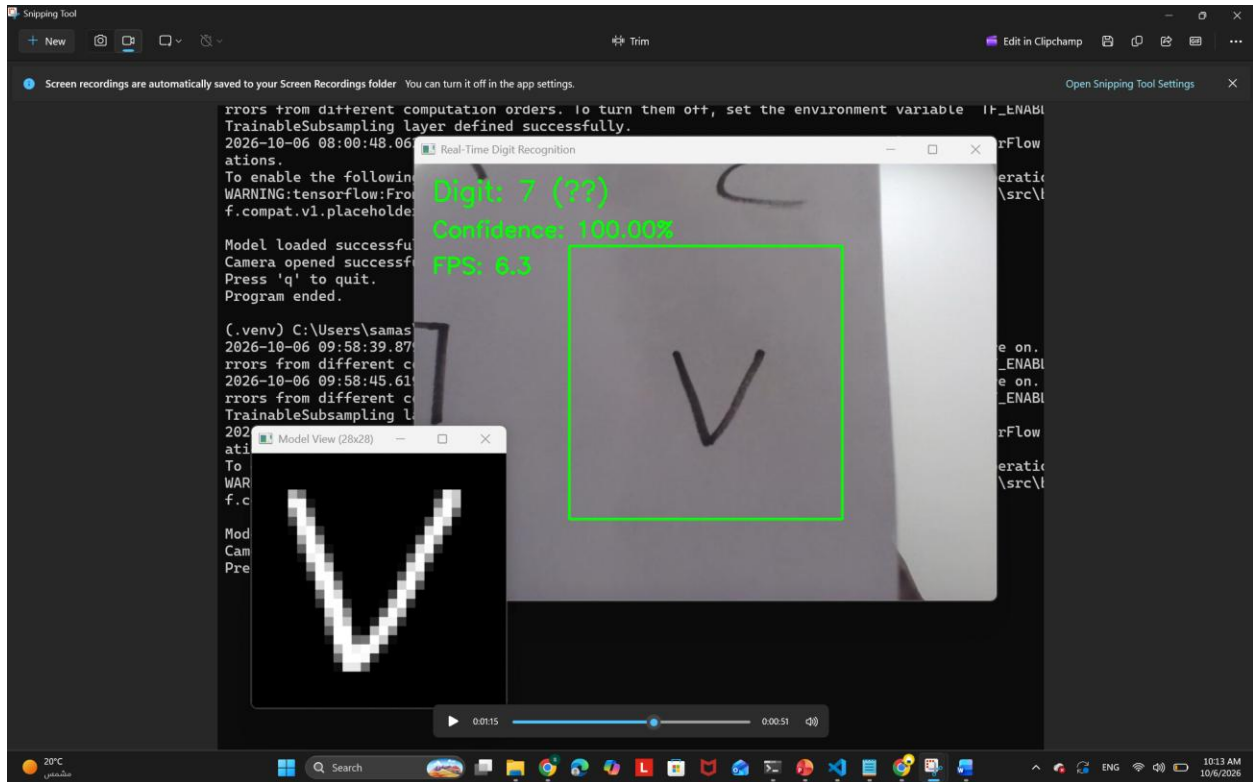
For the Arabic Bonus Part:

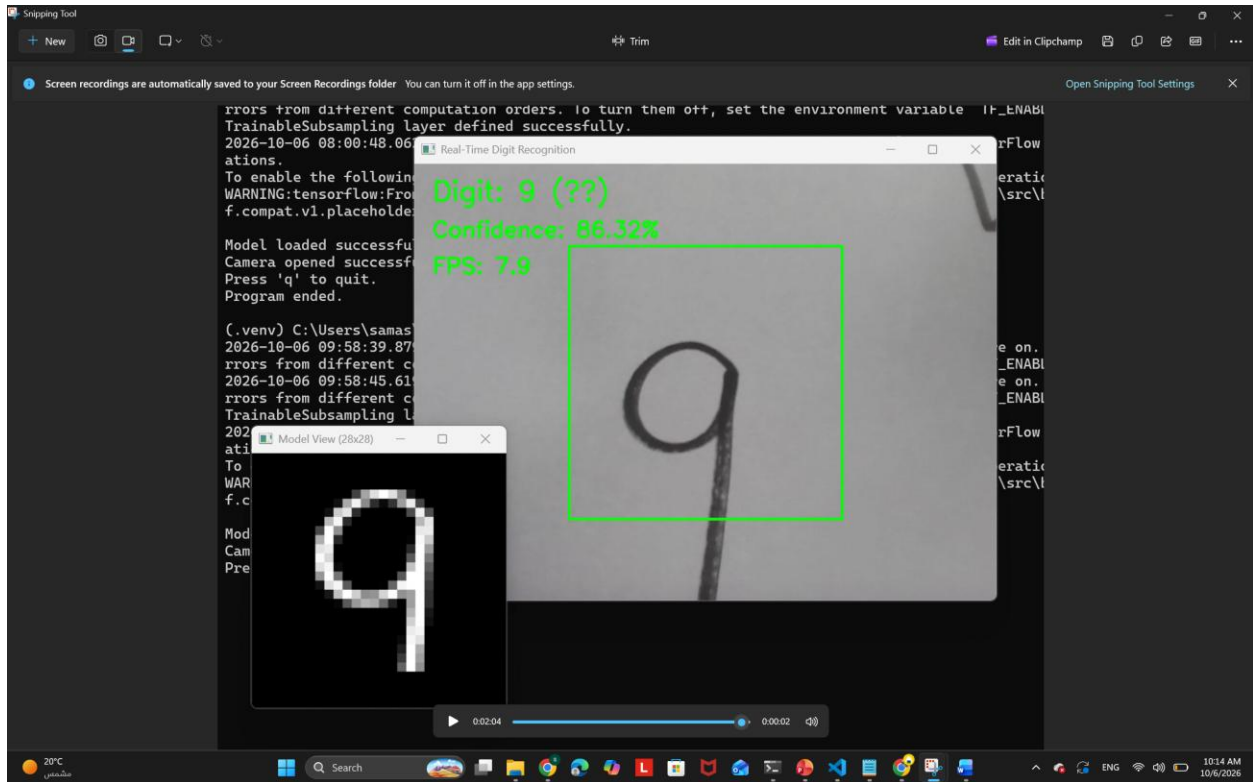












All numbers were detected, but for number 0, 9 it was confusing at first for the model when the number is far.

4. GitHub Repository.

<https://github.com/Nada44yehia/ELCT918.git>